

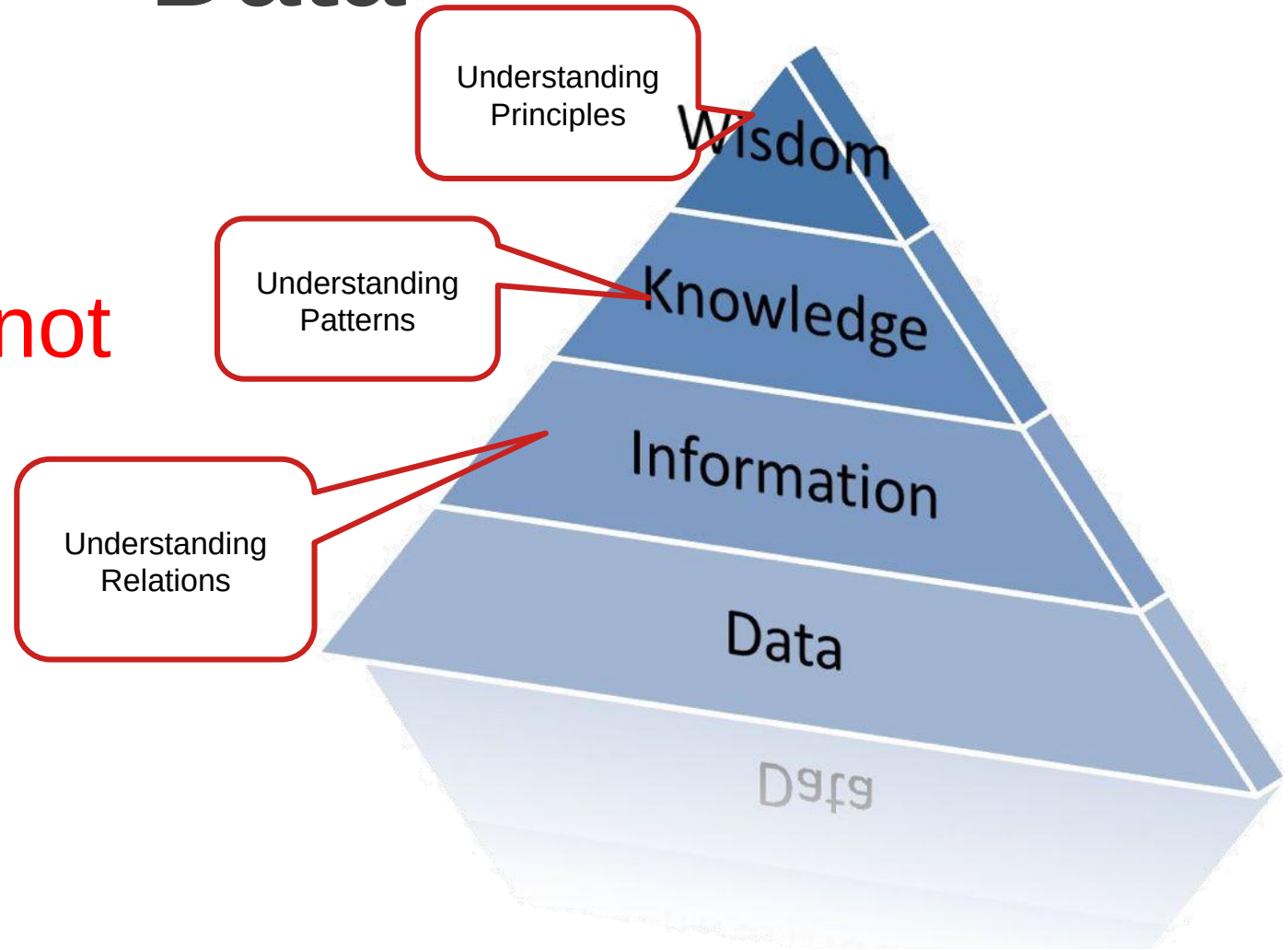
Database Management Systems

InterviewBit



Data

- A fact
- Raw, usable or **not**
- Simply exist



Type Data

- Structured Data
 - Strict format data like table data
- Semi Structured Data
 - Certain structure but not all have identical structure like share attributes. Format unstructured but content structured or vice versa. Example: XML, Json, CSV.
- Unstructured Data → its more related to Information Retrieval System
 - Limited indication the type of data like documents (Many Ekstenstion)

Type Data

```
'fuel_station_list': [  
  {  
    "access_days_time": "24 hours daily; restricted access 6pm-10pm",  
    "fuel_type_code": "CNG",  
    "groups_with_access_code": "Public - Credit card at all times",  
    "open_date": "1996-12-15",  
    "owner_type_code": "P",  
    "status_code": "E",  
    "station_name": "United Parcel Service",  
    "ng_vehicle_class": "MD",  
    "ng_fill_type_code": "Q",  
    "cards_accepted": [ "Master", "Visa" ],  
    "address": {  
      "latitude": 37.7739,  
      "longitude": -121.9695,  
      "city": "San Ramon",  
      "state": "CA",  
      "street_address": "4500 Norris Canyon Rd",  
      "zip": "94583"  
    },  
    "ng_psi": [[3000], [3600]]  
  },  
  ...  
]
```

Fuel Station

Cards Accepted

Address

PSI

Definitions

- **Data:** Known facts that can be recorded and have an implicit meaning.
- **Database:** A collection of related data.
- **Database Management System (DBMS):** A software package/ system to facilitate the creation and maintenance of a computerized database.
- **Database System:** The DBMS software together with the data itself. Sometimes, the applications are also included.
- **Mini-world:** Some part of the real world about which data is stored in a database. For example, student grades and transcripts at a university.

DBMS

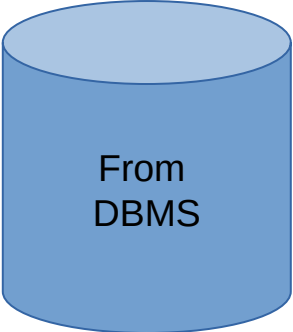
- **DBMS provides** efficient, reliable, convinient, safe multi-uses storage of and access to massive amount of persistent data
- **Database Management System** is a general purpose software system that facilitates the process of
 - defining,
 - constructing
 - manipulatingdatabase for various applications

Main function of DBMS

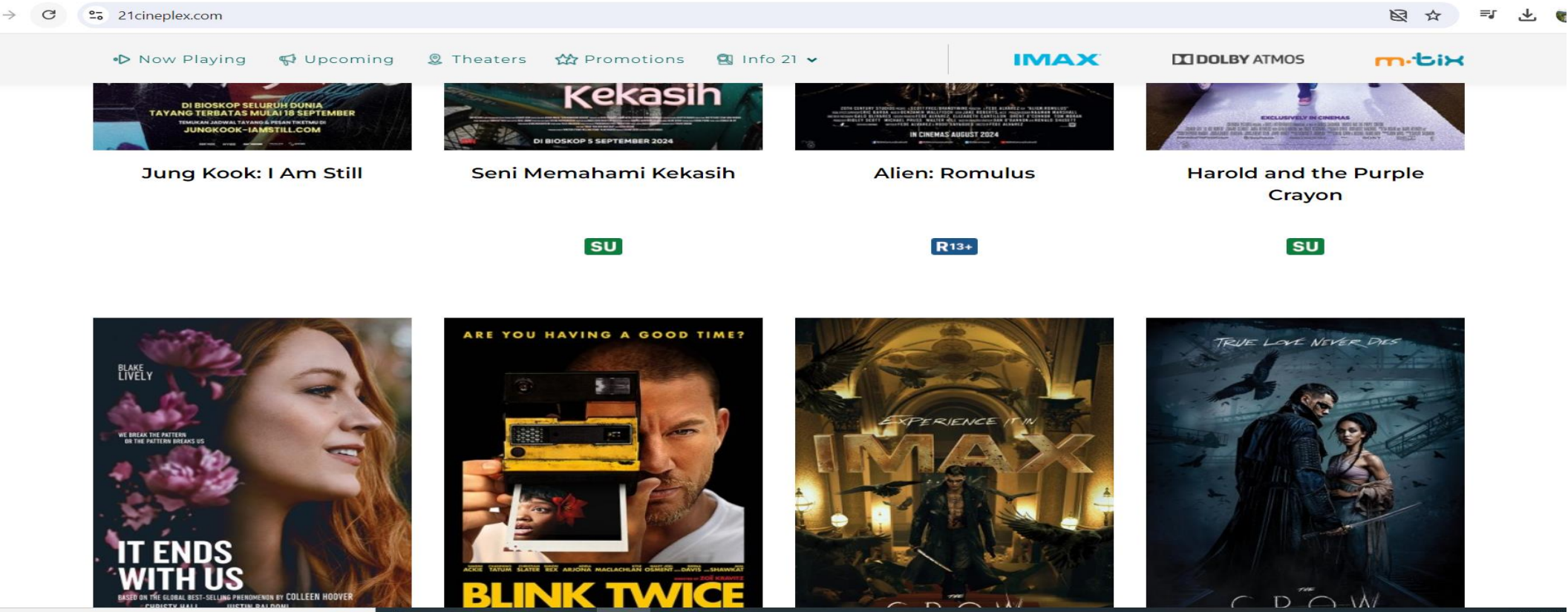
- **Defining a Database:**
 - Specifying Data Types, Structures, and Constraints
- **Constructing a Database:**
 - the Process of Storing the Data Itself on Some Storage Medium
- **Manipulating a Database:**
 - Function for Querying Specific Data in the Database and Updating the Database

Types of Databases and Database Applications

- Numeric and Textual Databases
- Multimedia Databases
- Geographic Information Systems (GIS)
- Data Warehouses
- Real-time and Active Databases
- Mobile Database
- Genome Data Management ..etc



Data



Types of Databases Text/numeric/multimedia ??

DBMS Tools

- MySQL,
- MSSQL,
- Oracle,
- OpenDatabase,
- MS.Access,
- PostGreSQL,



DBMS Tools

STRUCTURED VS UNSTRUCTURED VS SEMI-STRUCTURED DATA

	Structured data	Unstructured data	Semi-structured data
Formats	Tables, rows, columns	Text, images, audio, video	XML, JSON, CSV
Data model	Relational	None	Hierarchical/Graph
Common storages	Relational databases, traditional data warehouses	File systems, data lakes, cloud data warehouses	NoSQL databases
Data nature	Well-defined, fixed schema	Unpredictable, no schema	Loose schema
Analysis methods	SQL queries, data mining	NLP, image recognition, video analysis, text analysis, audio analysis, etc.	Query languages, data mining
Tools and technologies	Microsoft SQL Server, Oracle, MySQL	Amazon S3, Hadoop, Spark	MongoDB, Cassandra, Couchbase

Database Users

Actors on the scene

- **Database administrators:** responsible for authorizing access to the database, for coordinating and monitoring its use, acquiring software, and hardware resources, controlling its use and monitoring efficiency of operations.
- **Database Designers:** responsible to define the content, the structure, the constraints, and functions or transactions against the database. They must communicate with the end-users and understand their needs.
- **End-users:** they use the data for queries, reports and some of them actually update the database content.

Advantages of Using the Database Approach

1. Controlling redundancy in data storage and in development and maintenance efforts.
2. Sharing of data among multiple users.
3. Restricting unauthorized access to data.
4. Providing persistent storage for program Objects
5. Providing Storage Structures for efficient Query Processing
6. Providing backup and recovery services.
7. Providing multiple interfaces to different classes of users.
8. Representing complex relationships among data.
9. Enforcing integrity constraints on the database.
10. Drawing Inferences and Actions using rules

Additional Implications of Using the Database Approach

- Potential for enforcing standards: this is very crucial for the success of database applications in large organizations Standards refer to data item names, display formats, screens, report structures, meta-data (description of data) etc.
- Reduced application development time: incremental time to add each new application is reduced.

Additional Implications of Using the Database Approach

- Flexibility to change data structures: database structure may evolve as new requirements are defined.
- Availability of up-to-date information – very important for on-line transaction systems such as airline, hotel, car reservations.
- Economies of scale: by consolidating data and applications across departments wasteful overlap of resources and personnel can be avoided.

Historical Development of Database Technology

- **Early Database Applications:** The Hierarchical and Network Models were introduced in mid 1960's and dominated during the seventies. A bulk of the worldwide database processing still occurs using these models.
- **Relational Model based Systems:** The model that was originally introduced in 1970 was heavily researched and experimented with in IBM and the universities. Relational DBMS Products emerged in the 1980's.

Historical Development of Database Technology

- **Object-oriented applications:** OODBMSs were introduced in late 1980's and early 1990's to cater to the need of complex data processing in CAD and other applications. Their use has not taken off much.
- **Data on the Web and E-commerce Applications:** Web contains data in HTML (Hypertext markup language) with links among pages. This has given rise to a new set of applications and E-commerce is using new standards like XML (eXtended Markup Language).

Extending Database Capabilities

- **New functionality is being added to DBMSs in the following areas:**
 - Scientific Applications
 - Image Storage and Management
 - Audio and Video data management
 - Data Mining
 - Spatial data management
 - Time Series and Historical Data Management

The above gives rise to new research and development in incorporating new data types, complex data structures, new operations and storage and indexing schemes in database systems.

When not to use a DBMS

- **Main inhibitors (costs) of using a DBMS:**
 - High initial investment and possible need for additional hardware.
 - Overhead for providing generality, security, concurrency control, recovery, and integrity functions.
- **When a DBMS may be unnecessary:**
 - If the database and applications are simple, well defined, and not expected to change.
 - If there are stringent real-time requirements that may not be met because of DBMS overhead.
 - If access to data by multiple users is not required.

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When not to use a DBMS

- **When no DBMS may suffice:**
 - If the database system is not able to handle the complexity of data because of modeling limitations
 - If the database users need special operations not supported by the DBMS.

Key use DBMS

- **Data Model**
 - Categories of Data Models
 - History of Data Models
- **Data Schema**
- **Data Definition Language and Data Manipulation Language**

Data Model

- **Data Model:** A set of concepts to describe the *structure* of a database, and certain *constraints* that the database should obey.
- **Data Model Operations:** Operations for specifying database retrievals and updates by referring to the concepts of the data model. Operations on the data model may include *basic operations* and *user-defined operations*.

Categories of data models

- **Conceptual (high-level, semantic)** data models: Provide concepts that are close to the way many users *perceive* data. (Also called **entity-based** or **object-based** data models.)
- **Physical (low-level, internal)** data models: Provide concepts that describe details of how data is stored in the computer.
- **Implementation (representational)** data models: Provide concepts that fall between the above two, balancing user views with some computer storage details.

History of Data Models

- **Relational Model**: proposed in 1970 by E.F. Codd (IBM), first commercial system in 1981-82. Now in several commercial products (DB2, ORACLE, SQL Server, SYBASE, INFORMIX).
- **Network Model**: the first one to be implemented by Honeywell in 1964-65 (IDS System). Adopted heavily due to the support by CODASYL (CODASYL - DBTG report of 1971). Later implemented in a large variety of systems - IDMS (Cullinet - now CA), DMS 1100 (Unisys), IMAGE (H.P.), VAX -DBMS (Digital Equipment Corp.).
- **Hierarchical Data Model**: implemented in a joint effort by IBM and North American Rockwell around 1965. Resulted in the IMS family of systems. The most popular model. Other system based on this model: System 2k (SAS inc.)

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Model DBMS

STUDENT	Name	StudentNumber	Class	Major
	Smith	17	1	CS
	Brown	8	2	CS

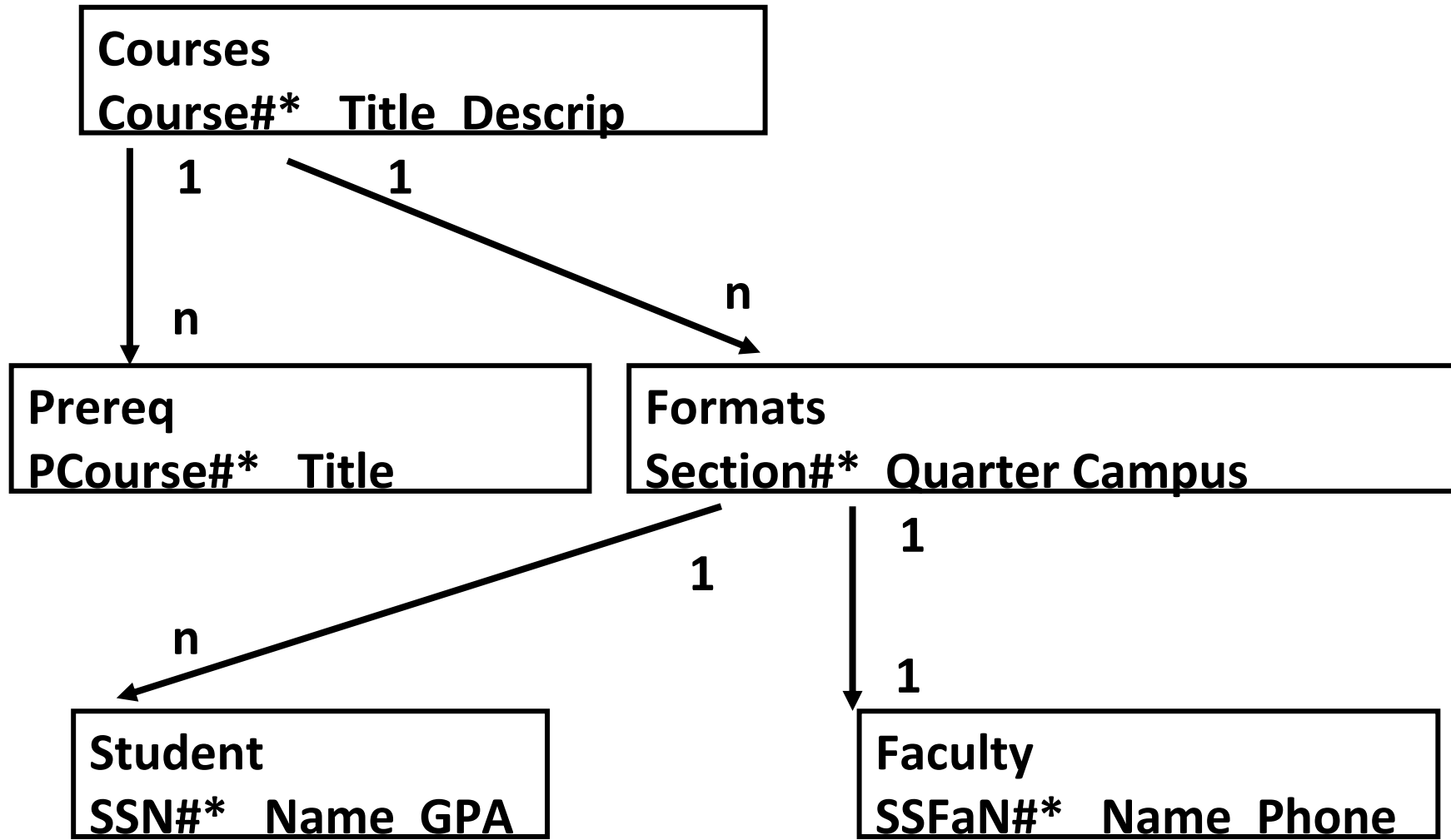
COURSE	CourseName	CourseNumber	CreditHours	Department
	Intro to Computer Science	CS1310	4	CS
	Data Structures	CS3320	4	CS
	Discrete Mathematics	MATH2410	3	MATH
	Database	CS3380	3	CS

SECTION	SectionIdentifier	CourseNumber	S
	85	MATH2410	
	92	CS1310	
	102	CS3320	
	112	MATH2410	
	119	CS1310	
	135	CS3380	

GRADE_REPORT	StudentNumber	SectionIdentifier	Grade
	17	112	B
	17	119	C
	8	85	A
	8	92	A
	8	102	B
	8	135	A

PREREQUISITE	CourseNumber	PrerequisiteNumber
	CS3380	CS3320
	CS3380	MATH2410
	CS3320	CS1310

ER Model DBMS



Schema

- **Database Schema:** The *description* of a database. Includes descriptions of the database structure and the constraints that should hold on the database.
- **Schema Construct:** A component of the schema or an object within the schema, e.g., STUDENT, COURSE.
- **Database Instance:** The **actual data stored** in a database at a *particular moment in time*. Also called **database state** (or **occurrence**).

DBMS Languages

- **Data Definition Language (DDL)**: Used by the DBA and database designers to specify the *conceptual schema* of a database. In many DBMSs, the DDL is also used to define internal and external schema (views). In some DBMSs, separate **storage definition language (SDL)** and **view definition language (VDL)** are used to define internal and external schema.

DBMS Languages

- **Data Manipulation Language (DML):** Used to specify database retrievals and updates.
 - DML commands (**data sublanguage**) can be *embedded* in a general-purpose programming language (**host language**), such as COBOL, C or an Assembly Language.
 - Alternatively, *stand-alone* DML commands can be applied directly (**query language**).

Database System Utilities

- To perform certain functions such as:
 - *Loading* data stored in files into a database. Includes data conversion tools.
 - *Backing up* the database periodically on tape.
 - *Reorganizing* database file structures.
 - *Report generation* utilities.
 - *Performance monitoring* utilities.
 - Other functions, such as *sorting*, *user monitoring*, *data compression*, etc.

Summary

- Data
- Database
- DBMS
- Advantages
- When Use or Not
- DBMS Tools



Barangsiapa yang berangkat menimba ilmu untuk mengamalkan ilmu,
niscaya ilmu yang sedikit pun akan bermanfaat baginya